

YIN Haoteng 殷浩騰

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EDUCATION

- **Purdue University** West Lafayette, IN
• **Ph.D. Computer Science (CIGP)**; Department of Computer Science, Advisor: Pan Li 2019 – 2025
- **Peking University** Beijing, China
• **MPhil Data Science**; Academy for Advanced Interdisciplinary Studies 2016 – 2019
- **Shandong University** Ji'nan, China
• **BEng Computer Science and Technology**; Taishan College* (Honors School) 2012 – 2016
• **BEcon Finance**; School of Economics 2014 – 2016
* Top-notch Undergraduate Training Program by Ministry of Education, China

RESEARCH INTERESTS

- Machine Learning and Computation on Complex Structured Data
Topics in Efficient and Scalable Learning on Graphs; LLMs and Foundation Models for Multimodal Structured Data; Trustworthy ML (Data Privacy, Ranking Fairness); Spatiotemporal Data Analysis.

SELECTED PUBLICATIONS (ASTERISK * INDICATES EQUAL CONTRIBUTIONS)

- **Haoteng Yin**, Rongzhe Wei, Eli Chien, Pan Li. Privately Learning from Graphs with Applications in Fine-tuning Large Pretrained Models. *Workshop on Statistical Foundations of LLMs and Foundation Models (SFLLM-NeurIPS'24)*
- **Haoteng Yin**, Jinha Kim, Prashant Mathur, Krishanu Sarker, Vidit Bansal. How to Talk to Language Models: Serialization Strategies for Structured Entity Matching. *Under Review*
- Rongzhe Wei, Eleonora Kreačić, Haoyu Wang, **Haoteng Yin**, Eli Chien, Vamsi K Potluru, Pan Li. On the Inherent Privacy Properties of Discrete Denoising Diffusion Models. *Transactions on Machine Learning Research (TMLR'24)*, Selected for Presenting at ICLR'25)
- **Haoteng Yin**, Beatrice Bevilacqua, Bruno Ribeiro, Pan Li. Enhancing Graph Representation Learning through Subgraph Strategies. *The 33rd International Joint Conference on Artificial Intelligence (IJCAI'24)*, Tutorial Track).
- Tianyi Zhang*, **Haoteng Yin***, Rongzhe Wei, Pan Li, Anshumali Shrivastava. Learning Scalable Structural Link Representations with Bloom Signatures. *Proceedings of the ACM Web Conference. (WWW'24)*
- Wei Xiang*, **Haoteng Yin***, He Wang, Xiaogang Jin. SocialCVAE: Predicting Pedestrian Trajectory via Interaction Conditioned Latents. *The 38th Annual AAAI Conference on Artificial Intelligence (AAAI'24)*.
- **Haoteng Yin**, Muhan Zhang, Jianguo Wang, Pan Li. SUREL+: Moving from Walks to Sets for Scalable Subgraph-based Graph Representation Learning. *Proceedings of the VLDB Endowment 16(11): 2939-2948*. DOI:10.14778/3611479.3611499 (**VLDB'23**)
- Prasita Mukherjee, **Haoteng Yin**[†]. OCTAL: Graph Representation Learning for LTL Model Checking. *The 10th International Workshop on Deep Learning on Graphs (DLG-KDD'23)*, Contributed Talk)
- Rongzhe Wei, **Haoteng Yin**, Junteng Jia, Austin R. Benson, Pan Li. Understanding Non-linearity in Graph Neural Networks from the Bayesian-Inference Perspective. *Advances in Neural Information Processing Systems 35 (NeurIPS'22)*
- **Haoteng Yin**, Muhan Zhang, Yanbang Wang, Jianguo Wang, Pan Li. Algorithm and System Co-design for Efficient Subgraph-based Graph Representation Learning. *Proceedings of the VLDB Endowment 15(11): 2788-2796*. DOI: 10.14778/3551793.3551831 (**VLDB'22**)
- Haorui Wang, **Haoteng Yin**, Muhan Zhang, Pan Li. Equivariant and Stable Positional Encoding for More Powerful Graph Neural Networks. *International Conference on Learning Representations (ICLR'22)*
- Bing Yu*, **Haoteng Yin***, Zhanxing Zhu. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. *The 27th International Joint Conference on Artificial Intelligence (IJCAI'18)*.
- **YIN Haoteng** and LIU Yang. Semantic analysis of spatial temporal trajectory in LBSNs. (in Chinese) *SCIENTIA SINICA Informationis 47(8), 1051-1065(2017)*. (**SMP'16 Best Paper Award**)

INVITED TALKS & PRESENTATIONS

- (2024) Privately Learning from Graphs with Applications in Fine-tuning Large Language Models. *M.I.N.D.S Group Seminar*, Hosted by Prof. Bruno Ribeiro, W. Lafayette, IN
- (2024) Privately Learning from Graphs with Applications in Fine-tuning Large Pretrained Models. *The 3rd RCAC Annual Cyberinfrastructure Symposium*, W. Lafayette, IN [**Best Poster Award, News**]
- (2024) Towards Representation Learning on (Sub)Graphs. *Purdue Database Group Seminar*, Hosted by Prof. Jianguo Wang, W. Lafayette, IN
- (2024) Algorithm and System Co-design for Scalable and Efficient Subgraph-based Graph Representation Learning. *Learning on Graphs & Geometry Reading Group*, Online [Link][Video]
- (2023) SUREL+: Moving from Walks to Sets for Scalable Subgraph-based Graph Representation Learning. *The 2nd RCAC Annual Cyberinfrastructure Symposium, CI + AI 2023*, W. Lafayette, IN
- (2023) Towards Scalable and Efficient Subgraph-based Graph Representation Learning. *Amazon A9 Machine Learning Reading Group*, Palo Alto, CA
- (2019) Spatio-Temporal Graph Prediction through Deep Graph Neural Networks. *The 3rd Peking University Graduate Student Forum on Advanced Interdisciplinary Studies*, Beijing, China [Video][**Best Oral Presentation Award**]

RESEARCH EXPERIENCE

- **Purdue University** West Lafayette, IN
Graduate Assistantships, under the supervision of Prof. Pan Li. Aug. 2020 - Current
 - **Differentially Private Learning from Graphs**
Design privacy-preserving learning frameworks for graph-structured data, with application in differential privately fine-tuning foundation models (e.g., Llama2-7B) on sensitive relational datasets.
 - **Efficient and Scalable Subgraph-based Graph Representation Learning (SGRL)**
Design novel frameworks SUREL/SUREL+ for subgraph-based GRL through algorithm and system co-design. A decoupled paradigm is proposed to sample and encode walk- and set-based subgraphs, which breaks the computational bottleneck and enables efficient training and inference on large-scale graphs for link, relation, and higher-order prediction tasks. SUREL/SUREL+ ranks in the Top 5 on Open Graph Benchmark datasets.
- **LIX, École Polytechnique & Inria** Palaiseau, France
Research Intern, under the supervision of Prof. Catuscia Palamidessi. Jun. 2018 - Sep. 2018
 - **Machine Learning Attacks to Location Privacy**
Develop an adversary toolkit that uses sequence models to capture users' movement patterns from perturbed trajectory data, which are gathered from location-based services and protected by a privacy mechanism, and then perform inference attacks to measure the vulnerability of the mechanism.
- **Center for Data Science, Peking University** Beijing, China
Graduate Research Assistant, under the supervision of Prof. Zhanxing Zhu Jan. 2017 - Aug. 2019
 - **Traffic Prediction with Deep Spatial Temporal Graph Neural Networks**
Design a deep learning framework STGCN to predict traffic time series, which coherently models the topology of road networks and captures periodic traffic patterns through spatiotemporal convolutional blocks. It has become a prominent model for network time series and is widely used in transportation research (4,400+ citations).

INDUSTRIAL EXPERIENCE

- **Amazon AGI** Bellevue, WA
Applied Scientist Intern May. 2024 - Aug. 2024
 - **Cross-Domain Entity Resolution with Graph-Enriched Language Models**
Mentor: Dr. Jinha Kim (Sr. Applied Scientist)
- **Amazon Search** Palo Alto, CA
Applied Scientist Intern Jun. 2023 - Sep. 2023
 - **Graph Foundation Models for Large-scale Text-attributed E-commerce Network**
Mentors: Shweta Garg (Sr. Applied Scientist) and Dr. Changhe Yuan (Principle Applied Scientist).

TECHNICAL SKILLS

- **Languages:** Chinese (native), English (professional working proficiency, OEPT certified)
- **Programming:** Proficient in Python/C, PyTorch, OpenMP, Transformers, Opacus, PyG, L^AT_EX, Git
Experienced in C++, PySpark, TensorFlow, Java, Julia, R, MATLAB

PROFESSIONAL SERVICES

- Conference Reviewer
 - ICLR'24-25, ICML'22-25, NeurIPS'22-24, WWW'22,24, WSDM'23, LoG'22-23, CISS'23, IEEE SPAWC'21
- Journal Reviewer
 - **Nature** Machine Intelligence; **ACM** Transactions on Spatial Algorithms and Systems; **IEEE** Transactions on Big Data, Cybernetics, Emerging Topics in Computing, Knowledge and Data Engineering, Information Forensics & Security, Intelligent Transportation Systems, Internet of Things Journal, Neural Networks and Learning Systems, Signal Processing
- Graduate Teaching Assistant
 - CS242 Introduction to Data Science - 19'Fall
 - CS243 Artificial Intelligence Basics - 24'Fall
 - CS38003 Python Programming - 20'Spring, 23'Spring
 - CS490-LDA Large Scale Data Analytics - 20'Fall
 - CS590-DE I&II Data Engineering - 20'Spring, 20'Summer, 21'Spring (*Lead)
 - CS590-NCDS Numerical Computing for Data Science - 21'Spring
 - CS593-CLG Computation & Learn on Graphs - 22'Fall (Guest Lecture: Build a Graph Learning Pipeline with PyG on Colab)
- Poster Session Judge - Purdue Spring Undergraduate Research Conference *Apr. 2024*
- Graduate Mentor - Purdue CS Graduate Student Board Mentor-Mentee Program *Aug. 2021, 2024*
- Student Volunteer - The International Machine Learning Society *Jul. 2018*
- Student Membership - AAI, ACM, SIAM

SELECTED HONORS & AWARDS

- The 3rd RCAC Cyberinfrastructure Symposium Best Poster Award (\$1,000) *Oct. 2024*
- Purdue Teaching Academy Graduate Teaching Award (2%) *Apr. 2024*
- AMiner AI 2000 Most Influential Scholar Award (Honorable Mention) in AAI/IJCAI *Mar. 2024*
- The Top Reviewer Award (8%) for NeurIPS'22 *Oct. 2022*
- Graduate School Summer Research Grants, Purdue University *Mar. 2021*
- Academic Excellence Scholarship, Peking University (~8%) *Oct. 2016, 2017, 2018*
- The 2nd DiDi-IEEE Elite Forum Extraordinary Potential Award (2/39) *May. 2018*
- École Polytechnique International Student Fellowship *Mar. 2018*
- The 5th National Conference of Social Media Processing the Best Paper Award (1/25) *Oct. 2016*
- Taishan College International Study Scholarship (CN¥33,000) *Jun. 2015*